

Presentation: Introduction to CSS & Selectors

1 What is CSS? (Brief Introduction)

- CSS stands for **Cascading Style Sheets**.
- It is used to **style HTML elements** and control the **layout** of web pages.
- Allows you to **separate content (HTML) from design (CSS)**.

2 Ways to Apply CSS

Inline CSS: Directly inside an HTML tag.

```
<p style="color: red;">This is red text.</p>
```

Internal CSS: Inside the `<style>` tag within `<head>`.

```
<style>
  p { color: blue; }
</style>
```

External CSS: A separate `.css` file linked using `<link>`.

```
<link rel="stylesheet" href="styles.css">
```

3 CSS Selectors (Targeting Elements)

Selectors define **which elements** you want to style.

Element Selector (Targets all instances of an element)

```
css
CopyEdit
p {
  color: blue;
}
```

Class Selector (Targets elements with a specific class)

```
css
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.highlight {
  background-color: yellow;
}
```

-

ID Selector (Targets a unique element with an ID)

```
css
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#header {
  font-size: 24px;
}
```

-

Universal Selector (Applies to all elements)

```
css
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* {
  margin: 0;
  padding: 0;
}
```

-

Group Selector (Applies styles to multiple elements at once)

```
css
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h1, h2, p {
  text-align: center;
}
```

-

4 CSS Properties: Colors & Backgrounds

Coloring Text

```
css
CopyEdit
p {
  color: red;
}
```

-

Background Colors

```
css
CopyEdit
body {
  background-color: #f4f4f4;
}
```

-

Background Images

```
css
CopyEdit
body {
  background-image: url('background.jpg');
  background-size: cover;
}
```

-

5 CSS Font Styling – Text Properties

In this section, we will cover various **CSS text properties** to style fonts, including **font family, size, weight, alignment, style, decoration, line height, letter spacing, word spacing, and text transformation**.

♦ 1. Changing Font Family

The **font-family** property is used to change the font style. It's a good practice to specify multiple fonts in case the first one isn't available.

```
css
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• p {
•   font-family: Arial, sans-serif; /* Arial is the primary
   font, sans-serif is fallback */
• }
```

💡 Common Font Families:

- **Serif:** Times New Roman, Georgia, Garamond
- **Sans-serif:** Arial, Helvetica, Verdana

- **Monospace:** Courier New, Consolas, Monaco
-

◆ 2. Setting Font Size

The `font-size` property controls the size of the text.

CSS

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- `h1 {`
- `font-size: 36px; /* Sets the font size to 36 pixels */`
- `}`

💡 Units for Font Size:

- `px` (Pixels) → Fixed size
 - `em` (Relative to parent element)
 - `rem` (Relative to the root `<html>` font size)
 - `%` (Percentage of the parent element's size)
 - `vw` (Viewport width, useful for responsiveness)
-

◆ 3. Font Weight (Boldness)

The `font-weight` property defines how **bold** or **thin** the text appears.

CSS

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- `p {`
- `font-weight: bold; /* Makes text bold */`
- `}`

💡 Other Values:

- `normal` (default)
- `bold`
- `lighter`
- `bolder`

- 100 (thin) to 900 (extra bold)

CSS

CopyEdit

- p {
 - font-weight: 300; /* Light font */
 - }
-

◆ 4. Text Alignment

The `text-align` property defines how text is positioned **inside its container**.

CSS

CopyEdit

- h1 {
- text-align: center; /* Centers the text */
- }

💡 Other Values:

- left (default for LTR languages)
 - right
 - center
-

◆ 5. Font Style (Italic or Normal)

The `font-style` property is used to make text **italic** or **oblique**.

CSS

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- p {
- font-style: italic;
- }

💡 Other Values:

- `normal` (default)
 - `italic`
 - `oblique` (slanted version of normal text)
-

◆ 6. Text Decoration (Underline, Overline, Strike-through)

The `text-decoration` property is used to style text with lines.

CSS

CopyEdit

- `a {`
- `text-decoration: none; /* Removes underline from links */`
- `}`
-
- `h2 {`
- `text-decoration: underline;`
- `}`
-
- `p {`
- `text-decoration: line-through; /* Strikethrough effect */`
- `}`

💡 Values:

- `none`
 - `underline`
 - `overline`
 - `line-through`
-

◆ 7. Line Height (Spacing Between Lines)

The `line-height` property controls the space **between lines of text**.

CSS

CopyEdit

- `p {`
- `line-height: 1.5; /* 1.5 times the font size */`
- `}`

💡 Best Practices:

- Use values like `1.5` or `2` instead of pixels for better readability.
-

♦ 8. Letter Spacing (Spacing Between Letters)

The `letter-spacing` property controls the **space between characters**.

CSS

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- `h1 {`
- `letter-spacing: 2px; /* Adds extra space between letters */`
- `}`

💡 Common Use Cases:

- Increase spacing for uppercase letters.
 - Improve readability for small text.
-

♦ 9. Word Spacing (Spacing Between Words)

The `word-spacing` property controls the **space between words**.

CSS

CopyEdit

- `p {`
- `word-spacing: 5px; /* Adds space between words */`
- `}`

💡 Usage Tip:

- Increasing word spacing can improve readability in large blocks of text.
-

◆ 10. Text Transformation (Uppercase, Lowercase, Capitalize)

The `text-transform` property is used to change the case of text.

CSS

CopyEdit

- `h1 {`
- `text-transform: uppercase; /* Converts text to uppercase */`
- `}`
-
- `h2 {`
- `text-transform: lowercase; /* Converts text to lowercase */`
- `}`
-
- `p {`
- `text-transform: capitalize; /* Capitalizes first letter of`
- `each word */`
- `}`

💡 Common Use Cases:

- `uppercase` for headings.
 - `capitalize` for titles.
 - `lowercase` for aesthetic styling.
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🎯 Summary of CSS Font Properties

Property	Description
<code>font-family</code>	Defines the typeface (Arial, Times, etc.)

<code>font-size</code>	Sets text size (px, em, rem, etc.)
<code>font-weight</code>	Defines text thickness (bold, 100-900)
<code>font-style</code>	Controls italics (<code>normal</code> , <code>italic</code> , <code>oblique</code>)
<code>text-align</code>	Aligns text (<code>left</code> , <code>center</code> , <code>right</code> , <code>justify</code>)
<code>text-decoration</code>	Adds lines (<code>underline</code> , <code>overline</code> , <code>line-through</code>)
<code>line-height</code>	Adjusts space between lines of text
<code>letter-spacing</code>	Modifies space between characters
<code>word-spacing</code>	Controls space between words
<code>text-transform</code>	Changes text case (uppercase, lowercase, capitalize)

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Requirements:

1. HTML Structure:

- Create a webpage with at least the following elements:
 - A header (`<h1>`) with the title of your webpage.
 - A navigation bar (`<nav>`) with 3-4 links (`<a>`).
 - At least two paragraphs (`<p>`) of content.
 - An image (`<img>` tag).
 - A footer (`<footer>`) with copyright text or a simple message.

2. CSS Styling:

- Use **external CSS** to style the page (link an external `.css` file to your HTML).
- Apply the following CSS selectors and properties:
 - **Element Selectors:** Style the `<h1>`, `<p>`, and `<footer>` elements.
 - **Class Selectors:** Use class selectors for any content you want to apply specific styles to, like highlighting a paragraph or styling a button.
 - **ID Selectors:** Style at least one unique element (e.g., the `<nav>` or `<footer>`) using an ID selector.
 - **Universal Selector:** Reset margins and paddings for all elements.
 - **Group Selectors:** Apply common styling (e.g., text color or font family) to headings and paragraphs.

3. Text Styling:

- Style text properties using the following:
 - Change the **font-family** of the body and headings.
 - Set the **font-size** for headings and paragraphs.

- Apply **font-weight** to emphasize certain parts of the text (e.g., bold headings or important points).
- Adjust the **text-align** property to center or right-align certain elements.
- Use **line-height** to improve the readability of your paragraphs.
- Add **text-decoration** to links (like removing the underline or adding a hover effect).
- Use **text-transform** to create uppercase or capitalize text.

4. Colors and Background:

- Set a **background color** for the body.
- Add a **background image** to either the body or header, making sure the image covers the full width or is positioned correctly.
- Style some text with **color** (e.g., change the color of links or headings).

5. Responsive Design:

- Use **viewport units** (vw, vh) or percentages to make your page more responsive to different screen sizes.
- Optionally, use **media queries** to change font sizes or layout for mobile devices.